

FORM 2
THE PATENTS ACT, 1970
(39 of 1970) & The Patents Rules 2003
COMPLETE SPECIFICATION
(SECTION 10 and Rule 13)

1. Title of the Invention:

**A control unit for evaluating the plausibility of data using a
generative intelligence model and a method thereof.**

2. Applicants:

a. Name: Bosch Global Software Technologies Private Limited

Nationality: INDIA

Address: 123, Industrial Layout, Hosur Road, Koramangala,
Bengaluru – 560095, Karnataka, India

b. Name: Robert Bosch GmbH

Nationality: GERMANY

Address: Postfach 300220, 0-70442, Stuttgart, Germany

Complete Specification:

The following specification describes and ascertains the nature of this invention and the manner in which it is to be performed.

5 **Field of the invention**

[0001] The invention is related to a control unit for evaluating the plausibility of data using a generative intelligence model and a method thereof.

Background of the invention

10 [0002] Wide range of training data concepts seems to be one of the most important components behind the zero-shot abilities of vision language foundation model (VLFM's). Nevertheless, it is virtually impossible to curate large amount of data for all the objects encountered in driving scenarios. Generative models come in handy to enhance the data quantitatively and qualitatively. These models only require handful
15 data to generate large high quality synthetic datasets. A major problem encountered while using generative VLFMs is, they are known to synthesize, hallucinated or deformed images at times, therefore making it important to assess the quality and plausibility of the generated images. Current SOTA metrics such as CLIP-I, DINO are not sufficient to benchmark the vision-linguistic capabilities of generative VLFMs, i.e., insufficient to
20 validate the plausibility of synthetic images. They mainly work on cosine similarity which in consequence cannot be used to provide high-level descriptions of a given image.

[0003] A technical paper having a title of "High-resolution image synthesis with latent diffusion models" discloses the similar concept. ☺

25 **Brief description of the accompanying drawings**

[0004] Figure 1 illustrates in a control unit for evaluating the plausibility of data using a generative intelligence model accordance with an embodiment of the invention; and

[0005] Figure 2 illustrates a flowchart of a method for evaluating the plausibility of
30 data using a generative intelligence model and a control unit in accordance with the present invention; and

[0005a] Figure 2a illustrates representations of the first multi-variate Gaussian distribution and a second multi-variate Gaussian distribution in accordance with an embodiment of the invention.

5

Detailed description of the embodiments

[0006] Figure 1 illustrates a control unit for evaluating the plausibility of data using a generative intelligence model in accordance with an embodiment of the invention. The control unit 10 comprises an image processing unit 14, a vision transformer 16, a feature generation unit 18. The control unit 10 curates a dataset 11a from a large set of images using atleast one curation technique for training the intelligence model 12 in a self-supervised approach . The control unit 10 comprehends the curated dataset 11a by focusing on each of the datapoint physical structure of the curated dataset 11a. The control unit 10 extracts feature maps related to each of a datapoints of the curated dataset 11a and generates a first mutli-variate gaussian distribution 13a based on an atleast one parameter. The control unit 10 feeds a synthetic dataset 11b to the intelligence model 12 and extracts the feature maps related to each of the datapoint of the synthetic dataset 11b for generating a second mutli-variate gaussian distribution 13b. The control unit 10 evaluates the plausibility of data present in the synthetic dataset 11b from the first and second generated mutli-variate gaussian distribution representations (13a & 13b).

[0007] Further the construction of the control unit and the working component of the control unit 10 is explained in detail. The control unit 10 and the operation of the different components connected to the control unit 10 is explained in detail. The methodology is deployed in the smart accident analysis unit 10 and the working of the control unit 10 is not a mere software program perse , but a means and function kind of arrangement where multiple hardware modules (a generative intelligence model, an image processing unit, a vision transformer, a feature generation unit, a filtering unit) are involved, each performing a particular task. The analysis unit 10 is an operating device that is used to process and to store the information received from atleast one source.

[0008] The control unit 10 is chosen from a group of analysis units comprising microprocessors, microcomputers, microcontrollers, digital signal processors, central

5 processing units, state machines, logic circuitries, one or more microchips or integrated
circuits interconnected using a parent board, hardwired logic, software stored by a
memory device and executed by a microprocessor, firmware, an application specific
integrated circuit (ASIC), and/or a field programmable gate array(FPGA), and/or any
component that operates on signals based on operational instructions. It is to be noted
10 that, the smart accident analysis unit 10 can be of any other kind, but not limited to the
above-mentioned devices which are known to a person skilled in the art.

[0009] The control unit 10 receives the large dataset (not shown) 10 from multiple data
sources. The large dataset comprises multiple images related to the vehicles and the
15 operating conditions of the vehicles. The curated dataset 11a as referred in this invention
is a sub-set of the large dataset that comprises the object of interest. For instance, the
object of the interest is a road lane marking. Each datapoint in the dataset (large or
curated 11a) is an image and in some parts of this invention, the data point is also
referred as an image. The control unit 10 considers/selects the images/datapoints from
20 the large dataset that comprises the object of interest for creating the curated dataset
11a. The curated dataset 11a thus formed is used to train the intelligence model 12.

[0011] The intelligence model 12 is an artificial intelligence model that is used to
evaluate the plausibility of the images/datapoints. According to one embodiment of the
25 invention, the intelligence model 12 is a vision Language foundation model (VILM). The
VILM 12 comprises the vision transformer for evaluating the plausibility of the synthetic
data 11b.

[0012] The comprehending of the curated dataset 11a in detail is attained by focusing
30 on a pixel-wise semantic relation of the objects of interest comprising a shape, a texture,
a color and the comprehend curated dataset 11a is used to train the vision transformer
of the intelligence model 12. The control unit 10 calculates two parameters for
generating the first mutli-variate gaussian distribution 13a and the two parameters are
a mean value and a standard deviation value. The mutli-variate gaussian distribution

5 13a/ 13b is a representation provided in the feature space based on the calculated parameters as mentioned above. However, it is to be noted that, the parameters are not restricted to the above-mentioned parameters, but can be any other parameters for obtaining/generating the Gaussian distribution.

10 **[0013]** The two parameters are calculated from a feature space created by the feature graphs of each of the datapoint of the curated dataset 11a. The control unit 10 performs a null hypothesis test of the features extracted with respect to the first mutli-variate gaussian distribution 13a and based on a comparison the datapoint is evaluated as plausible. The control unit 10 calculates a cosine similarity of each of the datapoint of the

15 synthetic dataset 11b and from the first gaussian distribution 13a and the second gaussian distribution representations 13b, the control unit 10 evaluates the plausibility of each of the datapoint of the synthetic dataset 11b.

[0014] When the calculated cosine similarity value of a particular datapoint of the

20 synthetic dataset 11b is more than a pre-defined value, then that datapoint is considered plausible by the control unit 10.

[0015] The control unit 10 one curated dataset 11a and a synthetic dataset 11b, the curated dataset 11a is used for initial training of the intelligence model 12 and the

25 synthetic dataset 11b is made plausible before training the intelligence model 12. The method of evaluating the datapoints of the synthetic dataset 11b is provided in this invention.

[0016] Figure 2 illustrates a flowchart of a method for evaluating the plausibility of data

30 using a generative intelligence model 12 and a control unit 10 in accordance with the present invention. The control unit 10 comprises an image processing unit 14, a vision transformer 16, a feature generation unit 18. In step S1, a dataset 11a from a large set of images is curated using atleast one curation technique for training the intelligence model 12 in a self-supervised approach. In step S2, the curated dataset 11a is

35 comprehended by focusing on each of the datapoint physical structure of the curated

5 dataset 11a. In step S3, feature maps related to each of a datapoints of the curated dataset 11a are extracted and a first mutli-variate gaussian distribution 13a is generated based on an atleast one parameter.

10 **[0017]** In step S4, a synthetic dataset 11b is fed to the intelligence model 12 and the feature maps related to each of the datapoint of the synthetic dataset 11b are extracted , for generating a second mutli-variate gaussian distribution 13b . In step S5, the plausibility of data is evaluated present in the synthetic dataset 11b from the first and second generated mutli-variate gaussian distribution representations (13a & 13b).

15 **[0018]** The method is explained in detail. The control unit 10 curates a data set 11a from a large dataset that is received from the multiple data sources (not shown). The curated dataset 11a is used to train the intelligence model 12 initially. The datapoints in the curated dataset 11a are pre-processed using the image processing unit 14. The features from the datapoints/images are extracted and the corresponding feature graphs are
20 generated. From the generated feature graphs, a feature space is created and two parameters (a mean value and a standard deviation value) are calculated to generate the first mutli-variate Gaussian distribution 13a. Each of the datapoint will have a feature map. The representation of the features are represented in the form of the Gaussian distribution 13a & 13b.

25

[0019] After this, the synthetic data set 11b that is used to train the intelligence model 12. The control unit 10 extracts the features of each of the datapoint of the synthetic dataset 11b and the corresponding feature maps are created. From the feature maps, a feature space is created ,where upon calculating the two parameters (a mean value and
30 a standard deviation value), the second multi-variate gaussian distribution 13b is represented in the corresponding feature space. The First multi-variate gaussian distribution 13a is related to the curated dataset 11a , wherein the second mutli-variate gaussian distribution 13b is related to the synthetic dataset 11b.

5 **[0020]** Then a cosine similarity of each of the datapoint of the synthetic dataset 11b is calculated. Based on a comparison between the first multi-variate gaussian distribution 13a and the second mutli-variate gaussian distribution 13b and the calculated cosine similarity, each of the datapoint in the synthetic dataset 11b is evaluated. The control unit 10 checks for the plausibility of each of the datapoint of the synthetic dataset 11b
10 based on the calculated cosine similarity. For instance, if the cosine similarity of the datapoint is more than the pre-defined value, then that particular datapoint is considered as plausible. Herein, the datapoint is an image.

[0021] The present invention uses the embedding space of VLFMs 12 such as vision
15 transformer 14 getting trained using self-supervised learning to evaluate the plausibility of synthetic dataset 11b images. Instead of evaluating plausibility using specific end-tasks like classification which requires labelled data, the present invention methodology analyzes the relationship between the representation of images generated using vision transformer 14. Embedding spaces generated using self-supervised learning can unveil
20 learned relationships that are difficult to measure in classical evaluation tasks. These tasks do not learn a probabilistic distribution but learns only the representation of the training data. Using these approaches results in high false negatives as they do not consider newly generated synthetic data 11b similar to the curated data 11a. The concept disclosed in this invention, overcomes this bottleneck by generating a multi-
25 variate Gaussian distribution (13a & 13b) which provides a basis for testing various hypothesis, allowing precise validation of plausibility.

[0022] The present invention aims to build a filtering mechanism which can determine the plausibility of the synthetic data 11b, given real images. The present invention
30 methodology can be viewed as a composition of four sequential processes as described below. The first one is with curating the real data samples/ curated dataset 11a datapoints of the object of interest from a vast collection of images. And then fine-tuning the generative intelligence model 12 which learns the distribution of real samples and then generates huge amount of novel(diverse) scenes of the object of interest using it.

5 This is followed by training the vision transformer 14 in self-supervised fashion which learns the semantic representation of the object of the interest present in the curated dataset 11a. The control unit 10 employs this vision transformer to generate a feature space of the curated dataset 11a and finally, the control unit 10 filters the images/datapoints using the trained vision transformer 14.

10

[0023] In the first stage of evaluating the plausibility, the control unit 14 curates' images containing object of interest from the large dataset of images using suitable methodology. This curated data 11a serves as the basis of the intelligence model 12 used for filtering the images. The control unit 10 then trains the intelligence model 12 using
15 the curated data 11a and employs it to synthesize new images. From the large dataset, the control unit 10 extracts an ample number of images containing the object of interest, enough to train to Vision Transformer 14 with Self-supervision of driving scenes object. The control unit 10 curates the images /dataset 11a using large scale vector similarity search techniques. Wherein one such similarity search technique is a FAISS technique.
20 The images are extracted such that the curated dataset 11a is a wide representation of shapes, textures, and colors of the object of the interest. The dataset 11a is pruned to remove any outliers present.

[0024] The pre-trained large scale latent-diffusion model (intelligence model) is fine-tuned it with the curated dataset 11a. Then the intelligence model 12 generates the new
25 images in various poses, styles, scenes, shapes, and viewpoints to ensure vast diversity in generated images while maintaining a high fidelity to the object of the interest. This allows to a create huge synthetic dataset 11b, which can boost the under-represented classes in the large dataset or even creating a new dataset for out-of-distribution objects
30 of interest.

[0025] The second stage involves, training a vision transformer 14. The curated data is used to train the vision transformer 14 in self-supervised fashion. This allows the intelligence model 12 to comprehend the images in detail, focusing on pixel-wise

5 semantic relation of objects i.e., their physical structure (shape, texture, color). The control unit 10 trains the Vision Transformer 14, using the curated dataset 11a and via the self-supervised learning.

10 **[0026]** The control unit 10 uses the open-source vision transformer 14 based foundational models to initialize the weights of the vision transformer 14, followed by self-supervised training. For a self-supervised training, the control unit 10 extracts the images from the curated dataset 11a, an anchor image, which is a reference sample, a positive image, , which is similar to the reference image, a negative image, which is dissimilar to the anchor image and use the triplet loss to train out the intelligence model 15 12. These images are fed to the vision transformer 14 to generate their features followed by calculation of triplet loss to train the intelligence model 12.

20 **[0027]** The triplet loss function is designed to ensure that the distance between features of the anchor image and the positive sample and the positive sample is smaller than the distance between the features of the anchor image and the negative sample at least by a margin α (a mean value). The loss, $\mathcal{L}$ is formulated as $\mathcal{L} = \max(d(\mathcal{M}(IA), \mathcal{M}(I+)) - d(\mathcal{M}(IA), \mathcal{M}(I-)) + \alpha, 0)$, wherein the IA = anchor image, $I+$ = Positive sample, $I-$ is the negative sample.

25 **[0028]** In the third stage, the control unit 10 generates the feature space as the multi-variate gaussian distribution. The control unit 10 utilizes the intelligence model 12 trained in the above disclosed stage, and the vision transformer 14 to generate features of individual images in the curated dataset 11a. The control unit 10 employs the image-wise self-attention mechanism as an integral component derived from the attention 30 blocks in the transformer architecture of the vision transformer. Specifically, the control unit 10 stores the maps from each self-attention block during inference of vision transformer 14 and stores them in an attention store (not shown) for each image. This composite attention map is then concatenated with embeddings generated by the vision

5 transformer 14 to further enhance the details of object of interest present in the image. This process is called generating feature space for the curated dataset 11a.

[0029] In the next stage which is referred as generating multi-variate gaussian distribution (the first and the second mutli-variate gaussian distributions), the generated features serves as the basis for the subsequent task of deriving the mean value and the standard deviation value (variance value) (μ, σ) for generating the first multi-variate Gaussian distribution, representing the overall image distribution. The control unit 10 generates features for both curated dataset (D) 11a and the synthetic dataset (S) 11b, derive ($\mu\mathcal{D}, \sigma\mathcal{D}$) and ($\mu\mathcal{S}, \sigma\mathcal{S}$) parameters. From these parameters, the control unit 10 generates the first and the second multi-variate Gaussian distributions (13a & 13b) $\mathcal{N}(\mu\mathcal{D}, \sigma\mathcal{D})$ and $\mathcal{N}(\mu\mathcal{S}, \sigma\mathcal{S})$ respectively. This allows us to represent the semantic structure of the object of interest in all the images as a probabilistic distribution of deterministic features, providing a basis for testing various hypotheses.

20 [0030] In the last stage, the filtering of the synthetic dataset 11b using the null hypothesis testing is performed. The generated features of each the image in synthetic dataset, 11b, $\mathcal{S}$, are then tested using the null hypothesis and the each image is considered plausible if and only if it lies with in the 95% confidence interval of the Gaussian distribution of the curated dataset 11a, $\mathcal{N}(\mu\mathcal{D}, \sigma\mathcal{D})$. This allows to efficiently use the embedding spaces of the vision transformer and benchmark the plausible images in a robust manner.

[0031] Figure 2a illustrates a visualization of the same is shown below – the green curve represents $\mathcal{N}(\mu\mathcal{D}, \sigma\mathcal{D})$ 13a, curated dataset 11a gaussian distribution and whereas the blue curve represents $\mathcal{N}(\mu\mathcal{S}, \sigma\mathcal{S})$ 13b, synthetic dataset 11b gaussian distribution. The dotted red lines represent the 95% confidence interval of $\mathcal{N}(\mu\mathcal{D}, \sigma\mathcal{D})$. The purple region shows the region of plausibility. The Confidence is calculated using the formula below.

Confidence= Hypothesis – Test($\mathcal{N}(\mu\mathcal{D}, \sigma\mathcal{D})$).

5

[0032] The control unit 10 allows the images i if *Confidence* is more than or equal to ≥ 0.95 . Please note that, the confidence parameter as mentioned in here is same as cosine similarity parameter.

10 **[0033]** With the above-mentioned methodology, one has developed a precise and robust system capable of evaluating the plausibility of a images synthesized using generative modelling. The filtering mechanism as mentioned in this present invention , computes spatial and semantic features of the generated image, and compare them with a probabilistic distribution of the real(curated) images 11a to generate a confidence of
15 plausibility, selecting only the images 11b with high plausibility score and rejecting images containing stringent hallucinations or deformations.

[0034] It should be understood that embodiments explained in the description above
20 are only illustrative and do not limit the scope of this invention. Many such embodiments and other modifications and changes in the embodiment explained in the description are envisaged. The scope of the invention is only limited by the scope of the claims.

5 **We Claim:**

1. A control unit (10) for evaluating the plausibility of data using a generative intelligence model (12) (a vision language foundation model- VLFM), said control unit (10) comprising an image processing unit (14), a vision transformer (16), a feature generation unit (18),
10 said control unit (10) configured to :
 - curate a dataset (11a) from a large set of images using atleast one curation technique for training the intelligence model in a self-supervised approach ;
 - comprehend the curated dataset (11a) by focusing on each of the datapoint physical structure of the curated dataset (11a);
 - 15 - extract feature maps related to each of a datapoints of the curated dataset (11a) by the feature extraction unit (18) and generate a first mutli-variate gaussian distribution (13a) based on an atleast one parameter ;
 - feed a synthetic dataset (11b) to the intelligence model (12) and extract the feature maps related to each of the datapoint of the synthetic dataset (11b)
 - 20 for generating a second mutli-variate gaussian distribution (13b);
 - evaluate the plausibility of data present in the synthetic dataset (11b) from the first and second generated mutli-variate gaussian distribution representations (13a & 13b) .
- 25 2. The control unit (10) as claimed in claim 1, the curated dataset (11a) comprises set of images which have an object of interest and is used for an initial training of the intelligence model (12).
3. The control unit (10) as claimed in claim 1, wherein the comprehending of the
30 curated dataset (11a) in detail is attained by focusing on a pixel-wise semantic relation of the objects of interest comprising a shape, a texture, a color and comprehend curated dataset (11a) is used to train the vision transformer of the intelligence model (12).

- 5 4. The control unit (10) as claimed in claim 1, wherein the control unit (10) adapted to calculate two parameters for generating the first mutli-variate gaussian distribution (13a) and the two parameters are a mean value and a standard deviation value.
- 10 5. The control unit (10) as claimed in claim 4, wherein the two parameters are calculated from a feature space created by the feature graphs of each of the datapoint of the curated dataset (11a).
- 15 6. The control unit (10) as claimed in claim 1, wherein the control unit (10) is adapted to perform a null hypothesis test of the features with respect to the first mutli-variate gaussian distribution (13a) and based on a comparison the datapoint is evaluated as plausible.
- 20 7. The control unit (10) as claimed in claim 6, wherein the control unit (10) calculates a cosine similarity of each of the datapoint of the synthetic dataset (11b) and from the first gaussian distribution (13a) and the second gaussian distribution representations (13b), the control unit (10) evaluates the plausibility of each of the datapoint of the synthetic dataset (11b).
- 25 8. The control unit (10) as claimed in claim 7, wherein when the calculated cosine similarity value of a particular datapoint of the synthetic dataset (11a) is more than a pre-defined value, then that datapoint is considered plausible by the control unit (10).
- 30 9. A method for evaluating the plausibility of data using a generative intelligence model (12) (a vision language foundation model- VLFM) by a control unit (10), the control unit (10) comprises an image processing unit (14), a vision transformer (16), a feature generation unit (18),
- 35 the method comprising the steps of :

- 5
- curating a dataset (11a) from a large set of images using atleast one curation technique for training the intelligence model (12) in a self-supervised approach ;
 - comprehending the curated dataset (11a) by focusing on each of the datapoint physical structure of the curated dataset (11a);
- 10
- extracting feature maps related to each of a datapoints of the curated dataset (11a) and generating a first mutli-variate gaussian distribution (13a) based on an atleast one parameter ;
 - feeding a synthetic dataset (11b) to the intelligence model (12) and extracting the feature maps related to each of the datapoint of the synthetic dataset (11b) for generating a second mutli-variate gaussian distribution (13b) ;
- 15
- evaluating the plausibility of data present in the synthetic dataset (11b) from the first and second generated mutli-variate gaussian distribution representations (13a & 13b).

20

Dated this 28th day of March 2025

(Digitally signed)

Siddharth Karkhanis

25

On-behalf of the Applicants (IN/PA-1195)

5 **Abstract**

A control unit for evaluating the plausibility of data using a generative intelligence model and a method thereof.

The control unit 10 comprises an image processing unit 14, a vision transformer 16, a feature generation unit 18. The control unit 10 curates a dataset 11a from a large set of
10 images for training the intelligence model 12. The control unit 10 comprehends the curated dataset 11a by focusing on each of the datapoint physical structure. The control unit 10 extracts feature maps related to each of a datapoints of the curated dataset 11a and generates a first mutli-variate gaussian distribution based on an atleast one parameter. The control unit 10 feeds a synthetic dataset 11b to the intelligence model
15 12 and extracts the feature maps related to each of the datapoint of the synthetic dataset 11b for generating a second mutli-variate gaussian distribution. The control unit 10 evaluates the plausibility of data of the synthetic dataset 11b from the first and second generated mutli-variate gaussian distribution.

20

(Figures1 & 2)